

EEE 419/591
Fall 2026
Additional Homework 1 – 591 Section

Solve the following additional problem to serve as your extra work for your enrollment in the 591 version of this class. Name your file hw1_591.py and submit it to the designated link on Canvas. Do not add the other problems to this code. If you upload a second attempt to Canvas and it renames your file, ignore the renaming.

In this HW, only limited AI is allowed. Examples of allowed prompts on this HW:

- What is a greatest common divisor of two numbers?
- How do I write an if statement in python? Give me an example
- How do I check if a number is integer in python?
- I got this error message. How do I interpret it?

Examples of prohibited prompts on this HW:

- Write me a code that calculates the greatest common divisor of a list of numbers in python
- I ran this code. [PASTE CODE HERE]. I got this error message. [PASTE THE ERROR MESSAGE HERE]. How do I fix my code?

Do not forget to cite your AI chat if you used any. Check the Syllabus document for details.

Problem 4: Manual Discrete Cosine Transform (DCT-II) Calculations

Calculate the Discrete Cosine Transform (type II) of a 1D signal array x_time and store the result in another array x_freq . The DCT-II is defined by:

$$X[k] = 2 \sum_{n=0}^{N-1} x[n] \cos \left[\frac{\pi}{N} \left(n + \frac{1}{2} \right) k \right], \quad k = 0, 1, \dots, N-1$$

Write a Python code that calculates the x_freq array using two distinct approaches:

1. Nested for loops
2. Matrix-vector multiplication (constructing the $N \times N$ DCT transformation matrix C)

Compare both of your manual implementations against the built-in `scipy.fftpack.dct` function (with `type=2`, `norm=None`). Your code should compute and print the maximum absolute error across all N values for both methods. Use any value for $N > 256$. A sample print out of your code could look like this:

Max error (loop vs SciPy DCT): $1.42e-13$

Max error (matrix vs SciPy DCT): $2.11e-13$

MANDATORY – Do not forget to add your mandatory citations as indicated on page 4 of the syllabus document. If you have not used any AI/references in your document, you still need to declare that as indicated in the syllabus document. This has 20% weight according to the rubric in the syllabus.